Software Engineering Models and Methods

TLA$^+$: [0/4] Intro

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Русский военный корабль

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Outline

Example

Labs

GCD

The Euclidean Algorithm

Outline

Example

Labs

Labs on TLA$^+$ | **Part 1**

Create a TLA$^+$ specification for a chosen algorithm and verify its properties by model-checking it.

Requirements:

- Before working on the specification, a student must get a confirmation from the lecturer if the problem is suitable for the lab task.
- The submitted lab has to be discussed with the lecturer to be evaluated.
- The student must be able to make requested changes to the specification.

Specify the same algorithm at a different abstraction level and provide a refinement mapping to the original one; prove an invariant of the algorithm.

Requirements:

- The same specification is extended in this lab.
- The submitted lab has to be discussed with the lecturer to be evaluated.
- The student must be able to make requested changes to the specification and the proof.
- Both the refinement and the proof are mandatory.

Labs

Example topics

The following is a list of example algorithms/protocols suitable for the lab, although other variants can (and are encouraged to) be proposed by students:

- A peer-to-peer chat program, considering the offline mode, maybe with presence support.
- The mDNS protocol (multicast DNS, used to discover devices in a local network).
- A chosen gossip protocol.
- A chosen distributed hash table.
- Data replication in a cluster.
- CRDTs.
- Distributed locks, snapshots, termination detection.

Lectures

1. Introduction.
2. Basics, syntax.
3. Abstractions and refinements.
4. Proof tactics and their use in TLA+.
5. Proofs on sets, naturals, temporal proofs.